TEXORA · LECTURE SHEET

Amino Acids and Proteins

The building blocks and their architecture: α-amino acids, zwitterions and isoelectric points, the peptide bond, the four levels of protein structure, fibrous proteins of textile fame — wool keratin and silk fibroin — denaturation, and the colour tests.

CHE-1205 · CHEMISTRY-II · TOPIC 7 · LEVEL 1 TERM 2

1. Amino acids — structure and families

Proteins are polymers of **α-amino acids**: a central carbon bearing –NH_2, –COOH, –H and a side chain R. Twenty standard R groups make the whole set; the chemistry of a protein is the sum of its R groups. Classify by the side chain:

FAMILY	MEMBERS / CHARACTER
Non-polar	Glycine (R = H), alanine, valine, leucine, isoleucine, proline, phenylalanine, methionine, tryptophan — shun water, hide inside folded proteins
Polar, uncharged	Serine, threonine, cysteine, asparagine, glutamine, tyrosine — hydrogen-bonding; cysteine's –SH forms disulphide bridges
Acidic	Aspartic and glutamic acids — an extra –COOH; negative at body pH
Basic	Lysine, arginine, histidine — extra basic groups; positive at body pH

Nine are **essential** — the body cannot synthesise them and diet must supply them (valine, leucine, isoleucine, threonine, methionine, phenylalanine, tryptophan, lysine, histidine). Except glycine, every α-amino acid is chiral, and nature uses the L forms almost exclusively — recall the chirality machinery of Topic 1.

2. Zwitterions and the isoelectric point

An amino acid carries both an acid and a base, so it neutralises itself: in the solid and near neutral pH it exists as a **zwitterion**, $^{+}H_3N–CHR–COO^{-}$ — which is why amino acids are crystalline, high-melting and water-soluble like salts, not like ordinary organic molecules. At low pH it is a cation ($^{+}H_3N–CHR–COOH$); at high pH an anion ($H_2N–CHR–COO^{-}$). The pH at which the net charge is zero — the molecule will not migrate in an electric field — is the **isoelectric point** pI, the mean of the two relevant pKa values:

WORKED EXAMPLES

Glycine (pKa_1 = 2.34, pKa_2 = 9.60): pI = (2.34 + 9.60)/2 = **5.97**.
Aspartic acid (extra –COOH: pKa 1.88, 3.65, 9.60): the two lowest bracket the neutral form: pI = (1.88 + 3.65)/2 = **2.77**.
Lysine (extra $–NH_2$: pKa 2.18, 8.95, 10.53): the two highest: pI = (8.95 + 10.53)/2 = **9.74**.
Rule: for side-chain acids average the two lowest pKa values; for side-chain bases, the two highest. At its pI a protein is least soluble — the principle behind isoelectric precipitation in dairy and dyehouse effluent treatment alike.

3. The peptide bond and the four levels of structure

Amino acids polymerise by condensation: the –COOH of one and $–NH_2$ of the next eliminate water, forming the **peptide bond** –CO–NH– — planar and partially double-bonded, so the backbone is a chain of rigid plates. A chain is read from its free N-terminus; Gly–Ala and Ala–Gly are different molecules. Structure then stacks in four levels:

LEVEL	WHAT IT IS	HELD BY
Primary	The amino-acid sequence (Sanger's sequencing of insulin, 51 residues in two chains, was the landmark proof that proteins have exact sequences)	Peptide bonds (covalent)
Secondary	Local folding: α-helix (3.6 residues per turn) and β-pleated sheet	Hydrogen bonds between backbone C=O and N–H
Tertiary	The whole chain folded into its working 3D shape	R-group interactions: hydrophobic effect, H-bonds, ionic links, disulphide bridges
Quaternary	Several chains assembled (haemoglobin: four subunits)	The same non-covalent forces, between chains

Fibrous vs globular: fibrous proteins are structural — keratin (α-helical) of wool and hair, collagen's triple helix, silk fibroin's β-sheets. Globular proteins fold into compact balls and do chemistry: enzymes, antibodies, haemoglobin. For a textile student the fibrous pair is home ground: wool's spring and silk's sheen are secondary structures you can touch.

4. Denaturation

Heat, strong acid or alkali, urea, heavy-metal salts or violent agitation unravel a protein's secondary, tertiary and quaternary structure — **denaturation**. The peptide bonds (primary structure) survive; the

function does not: egg white sets, enzymes die, milk curdles at its isoelectric point. Wool chemistry is applied denaturation control: the **disulphide bridges** of keratin give wool its elasticity; permanent-waving and wool setting work by reducing S–S bonds, reshaping the fibre, and re-oxidising the bridges in the new shape — and harsh chlorination or alkali in wet processing damages exactly these bonds, which is why wool demands gentle scouring (compare TCC-I).

5. Colour tests

TEST	REAGENT / OBSERVATION	DETECTS
Ninhydrin	Heat with ninhydrin → deep purple (Ruhemann's purple); proline, a secondary amine, gives yellow instead	Free α-amino acids — also develops latent fingerprints
Biuret	Dilute $CuSO_4$ in alkali → violet	Two or more peptide bonds — proteins and peptides, *not* free amino acids
Xanthoproteic	Conc. HNO_3 → yellow, turning orange with alkali	Aromatic side chains (tyrosine, tryptophan, phenylalanine)

6. Key points

- α-amino acids differ only in R; classify R as non-polar, polar, acidic or basic; nine are essential.
- Zwitterion form explains salt-like properties; pI = mean of the bracketing pKa values (acidic: two lowest; basic: two highest); zero net charge and minimum solubility at pI.
- Peptide bond: planar, covalent; structure levels run primary → secondary (H-bonds) → tertiary (R-group forces, S–S bridges) → quaternary.
- Denaturation destroys levels 2–4, never the sequence; keratin's disulphide bridges are the working chemistry of wool setting and wool damage.
- Ninhydrin flags free amino acids (purple; proline yellow); biuret needs ≥ 2 peptide bonds; xanthoproteic flags aromatic side chains.

7. Practice

Answer in your own words; brief model answers follow.

1. Calculate the isoelectric point of glycine (pKa_1 = 2.34, pKa_2 = 9.60), and state glycine's predominant form at pH 2 and at pH 11.

2. A solution gives a violet biuret test but no purple with ninhydrin on brief warming. What does the pair of results tell you?
3. Boiling an egg irreversibly sets the white, yet the protein's amino-acid sequence is unchanged. Name the phenomenon and state exactly which structural levels are destroyed and which survives.
4. Why does proline give a yellow colour with ninhydrin while other α-amino acids give purple?
5. Explain, from keratin's structure, how a permanent wave works and why strongly alkaline scouring damages wool.
6. At its isoelectric point a protein shows minimum solubility and does not migrate in electrophoresis. Explain both facts with one idea.

Answer key

1. pI = (2.34 + 9.60)/2 = 5.97. At pH 2 (below both pKa values) the carboxyl is protonated: $^{+}H_3N–CH_2–COOH$, a cation. At pH 11 (above both) the ammonium is deprotonated: $H_2N–CH_2–COO^{-}$, an anion. Between them sits the zwitterion.
2. The sample contains peptide chains (biuret needs at least two peptide bonds) but essentially no free α-amino acids — a protein or long peptide, undigested. Hydrolyse it and the ninhydrin test would turn strongly positive as free amino acids appear.
3. Denaturation. Secondary, tertiary (and any quaternary) structure — held by hydrogen bonds and other non-covalent forces — is destroyed; the primary structure, the covalent peptide-bonded sequence, survives intact. Function follows shape, so function is lost without any bond in the backbone breaking.
4. Proline's amino group is secondary (its nitrogen is inside a ring, bonded to the α-carbon twice over — an imino acid). Ninhydrin's purple product requires the imine chemistry of a primary α-amino group; proline's pathway stops at a yellow iminium product.
5. Wool keratin is α-helical protein cross-linked by cystine disulphide bridges between chains. A perm uses a reducing agent to break S–S bridges, curlers impose a new shape, and an oxidiser reforms the bridges there — the curl is stored in covalent cross-links. Strong alkali attacks the same disulphide bonds (and hydrolyses peptide links), so over-harsh scouring permanently weakens and yellows wool.
6. Net charge is zero at the pI. With no net charge, molecules are not pulled to either electrode (no electrophoretic migration), and with no charge repulsion keeping them apart, they aggregate and precipitate — minimum solubility. One fact, both observations.
